

The Chip That Turns AI MODELS INTO SILICON

The AI industry has spent years scaling models through brute-force computing. HC1 takes a different path entirely, collapsing hardware and software into a single architectural layer.

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A radical compute-in-memory architecture is collapsing the boundary between hardware and AI models, eliminating data movement and redefining performance-per-watt. The HC1 by Taalas is not just faster, it signals a fundamental shift in how chips are conceived, built, and optimised for the next era of AI inference.

The unveiling of the HC1 is not merely another addition to the crowded AI chip market—it represents a fundamental rethinking of how artificial intelligence workloads are executed. HC1 addresses inefficiencies in current AI infrastructure by targeting escalating costs, rising energy demands, and architectural bottlenecks that are beginning to constrain large-scale AI deployment.

At its core, the HC1 introduces a new category of compute hardware in which AI models are no longer software artefacts running on general-purpose processors. Instead, they are physically instantiated into silicon. This shift effectively turns the chip into the model itself, collapsing the distinction between hardware and software while enabling performance levels that

exceed current inference benchmarks.

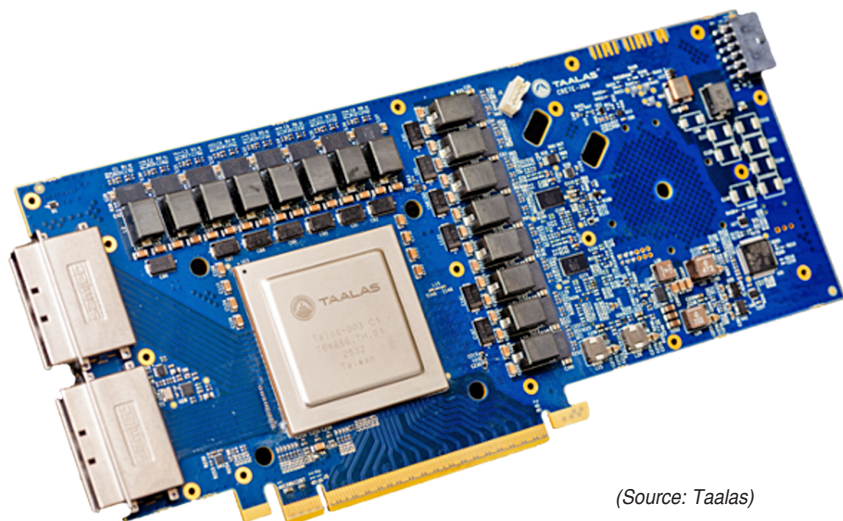
The company claims the HC1 achieves throughput of 17,000 tokens per second on Llama 3.1 8B—an order-of-magnitude leap over existing systems. Crucially, this performance is delivered without reliance on high-bandwidth memory (HBM), liquid-cooling systems, or hyperscale infrastructure. A fully configured server equipped with ten HC1 cards operates at approximately 2.5kW using standard air cooling, dramatically lowering the barrier to deployment.

The infrastructure crisis driving innovation

The HC1 emerges against the backdrop of a growing crisis in AI infrastructure, where traditional system architectures are struggling to keep pace with demand. The central issue lies in the ‘memory wall’—a long-standing limitation in computing where data transfer between memory and processing units becomes the dominant bottleneck.

In conventional AI systems, compute units such as GPUs are physically separated from memory. Each inference operation requires continuous movement of model weights and intermediate data across this boundary. This process is not only time-consuming but also highly energy-intensive, with estimates suggesting that 80-90% of total energy consumption is spent on data movement rather than arithmetic itself, a figure increasingly validated by real-world data centre telemetry.

These inefficiencies manifest as increased latency, excessive heat generation, and the need for complex cooling and infrastructure systems. Even the most advanced chips today, delivering fewer



(Source: Taalas)